Ch13 溶液

-Solute (n.)溶质

-Solvent (n.)溶剂

-The ability of substances to form solutions depends on 取决于

1.natural tendency toward mixing

2.intermolecular forces

-Mixing causes more randomness in the position of the molecules,

Increasing a thermodynamic quantity called entropy.

-伴随着 mixing 的 entropy ↑，溶液的形成 ↑

-Any intermolecular force of attraction (Chapter 11)(Dispersion, Dipole-dipole, Hydrogen bond, Ion-dipole) can be the attraction between solute and solvent molecules.

-**supersaturated solution**: crystallization can usually be stimulated by adding a "seed crystal" or scratching the side of the flask.

Factors That Affect Solubility

1. Solute–solvent Interactions

the solute–solvent interaction ↑, the solubility of a solute in that solvent ↑

The larger the gas个, the more soluble个 it will be in water. (?范德华力?)

miscible 混溶的，immiscible 不混溶的

2.Pressure (for gaseous solutes)

Henry's Law

The solubility of a gas is proportional to the partial pressure of the gas above the solution.

$$S_g = kP_g$$

3.Temperature

T ↑，大多数固体溶解度 ↑

T ↑，所有气体溶解度 ↓

-Units of Concentration

Mass percentage $\text{Mass \% of component} = \frac{\text{mass of component in soln}}{\text{total mass of soln}} \times 100$;

Parts per million (ppm) 10^6

Parts per **billion 十亿** (ppb) 10^9

$$\text{ppm of component} = \frac{\text{mass of component in soln}}{\text{total mass of soln}} \times 10^6$$

Mole fraction;

Molarity $\text{Molarity} = \frac{\text{moles of solute}}{\text{liters of soln}}$

Molality $\text{Molality} = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$

当水是溶剂时，稀溶液具有相似的 molarity 和 molality; molality 不随温度变化（质量不变）; molarity 随温度变化（体积变化）。

-Colligative properties depend only on the quantity, not on the identity of the solute particles. 蒸汽压 ↓;沸点 ↑;凝固点 ↓;渗透压

-the vapor pressure of a solution < the pure solvent.

-Raoult's Law

挥发性溶剂在溶液上的蒸气压是溶剂的摩尔分数乘以纯溶剂蒸气压的乘积。

$$P_{\text{solution}} = X_{\text{solvent}} P^{\circ}_{\text{solvent}}$$

-Boiling-Point **Elevation 升高** and Freezing-Point Depression(using the van't Hoff factor)

Vapor pressure ↓，沸点 ↑，凝固点(熔点) ↓

$$\Delta T_b = T_b(\text{solution}) - T_b(\text{solvent}) = iK_b m$$

$$\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent}) = -iK_f m$$

The van't Hoff Factor (i): 解离到溶液中的粒子种数

-The net movement of solvent molecules from solution of low to high concentration across a semipermeable membrane is **osmosis 渗透**. The applied pressure to stop it is **osmotic pressure 渗透压**.

semipermeable membranes 半透膜

-Osmotic pressure is a **colligative(adj.)**碰撞的 property. $\Pi = i\left(\frac{n}{V}\right)RT = iMRT$

-Isotonic solutions 等渗溶液

Hypotonic solution 低渗溶液

Hypertonic solution 高渗溶液

-colloids 胶体

Tyndall effect 丁达尔效应

hydrophilic 亲水的(极性)；hydrophobic 疏水的(非极性)

-Stabilizing Colloids by Adsorption

离子可以粘附在其他疏水胶体的表面上。

这允许它与水溶液相互作用。

胶体可以帮助脂肪和油在水溶液中乳化。

emulsifier 乳化剂会导致通常不溶于溶剂的东西这样做。

-Brownian Motion 布朗运动

胶体由于与小得多的溶剂发生多次碰撞而引起的运动。

Ch14 化学动力学

-Factors that Affect Reaction Rates

1. Physical state of the reactants

反应物碰撞准备(???) ↑，反应速度 ↑

反应速度 Homogeneous reaction > Heterogeneous reaction.

Heterogeneous reactions 中，固体反应物表面积个，反应速率个

2. Reactant concentrations

反应物浓度个，反应速率个

原因：分子数个，碰撞数个

3. Reaction temperature

温度个，反应速率个

温度个，分子运动速度个，碰撞数个，碰撞中分子拥有的能量个，分子动能 ↑

4. Presence of a catalyst

催化剂影响碰撞的种类，改变机制（从反应物到产物的途径中的单个反应

-Rate is dependent on **stoichiometry 化学计量数** $\text{Rate} = -\frac{1}{2}\frac{\Delta[O_3]}{\Delta t} = \frac{1}{3}\frac{\Delta[O_2]}{\Delta t}$

-First Order Reactions: The rates depend only on one reactant to the first power.

The rate law becomes: Rate = k[A]

rate = k[A]

↘ k[A] = − Δ [A] / Δ t

rate = − Δ [A] / Δ t

Rearrange: ln[A] = −kt + ln [A]$_0$

→从图中找 k，斜率为 k

E.g. $CH_3NC \rightarrow CH_3CN$, rate = k [$CH_3NC$]

(?A 为反应物?)

Half-life 半衰期: ln 2 = k t½ or t½ = 0.693/k

-Second Order Reactions

Def.: The rates depend only on a reactant to the second power.

The rate law becomes: Rate = k[A]^2

k [A]$_2$ = −Δ[A] / Δ t

Using calculus: 1/[A] = 1/[A]$_0$ + k t

E.g. NO2→NO+(1/2)O_2

Its second order rate = k[NO2]^2

Half-life 半衰期: t½ = 1 / (k [A]$_0$)

So half-life is a concentration dependent quantity for second order reactions!

-Zero Order Reactions

rate is independent of the concentration of the reactant ; Rate = k

These reactions are linear in concentration.

-Factors That Affect Reaction Rate

1. Temperature

温度每+10℃，Rate constant×2

2. Frequency of collisions

分子数个 / 温度个，分子速率个

→碰撞数个，反应数个，反应速率个

3. Orientation of molecules

Bonds must be broken and made and atoms need to be in proper positions.

4. Energy needed for the reaction to take place (activation energy)

Def.: The minimum energy needed for a reaction to take place

-Transition State (Activated Complex): The organization of the atoms at this highest energy state. ;在最高能量状态下，形成 transition state 在此之后，反应可以是吸热或放热

Activation energy: The energy needed to form

transition state

-能量个，拥有反应发生所需的能量的分子数个

- $\ln\frac{k_1}{k_2}=\frac{E_a}{R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$ E_a: 活化能 T: 温度

-Mechanism

Def.: a series of stepwise reactions that show how reactants become products

-Elementary Reaction (又: Elementary Process)

Def.: Each of these processes (discrete steps) that reactions (may) occurs through.

(反应可以同时发生，也可以通过几个离散的步骤发生。这些过程中的每一个都被称为基本反应或基本过程。)

-Molecularity

The molecularity of an elementary reaction ↓可知

The number of molecules that are involved in that step of the mechanism

-Termolecular steps 三分子步骤

require three molecules to **simultaneously(adv.)同时地** collide

速率 < unimolecular or bimolecular steps

-the rate-determining step 决速步

-What is Required of a Plausible(adj.)合理的 Mechanism?

The rate law 必须能够从 the rate-determining step 中设计出来; 当所有步骤相加时，必须可以获得 The stoichiometry; 每一步都必须平衡，就像任何方程式一样;所有 intermediates 都 被制成并被用完;任何 catalyst 催化剂都被使用并再生

-Catalysts 催化剂: Activation energy is lowered, allowing equilibrium to be established at lower temperatures

1.Homogeneous catalysts 均相催化剂

The reactants and catalyst are in the same phase; reactants and catalyst are dissolved in the same solvent

2.Heterogeneous catalysts 非均相催化剂

Often: gases are passed over a solid catalyst;

The **adsorption(n.)吸附** of the reactants is often the rate-determining step.

3.Enzymes 酶

Active site: region where the reactants attach

Substrates 底物: The reactants are referred to as substrates

Ch15 化学平衡

-For the forward reaction: $N_2O_4(g) \rightarrow 2NO_2(g)$

The rate law is: $Rate_f = k_f [N_2O_4]$

For the reverse reaction: $2NO_2(g) \rightarrow N_2O_4(g)$

The rate law is: $Rate_r = k_r [NO_2]^2$

$Rate_f = Rate_r$

-the equilibrium constant (Keq)

$K_{eq} = \frac{[NO_2]^2}{[N_2O_4]} = \frac{k_f}{k_r} = \text{a constant}$

因 k 与反应物浓度为乘法关系，所以此处 $\frac{[NO_2]^2}{[N_2O_4]}$ 为 $\frac{k_f}{k_r}$ 反比 !

-the Haber Process: the industrial preparation of ammonia 氨的工业制备 $K_c=\frac{[NH_3]^2}{[N_2][H_2]^3}$

-For gases, $K_p = K_c(RT)^{\Delta n}$,where $\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$

-K>>1 → favors products; K<<1...

-Consecutive(adj.)连续的,连贯的 Equilibria

$K_3 = K_1 \times K_2$

-the concentration of a pure substance is 1

-Homogeneous equilibria 单相平衡: all reactants and products are in the same phase.

-Heterogeneous equilibria 多相平衡: something in the equilibrium is in a different phase.

-reaction quotient (Q): the current conditions

If Q < K, the reaction is proceed to products. ;...

-LeChâtelier's Principle 勒夏特列原理

counteract(v.)抵抗

Ch16 酸碱平衡

-Arrhenius 阿伦尼乌斯: Acid 酸性物质 是 溶解到水中，会增加 H^+的浓度 的物质; **Base 碱性物质** ...

-Brønsted – Lowry 布朗斯特-劳里: An acid is a proton donor. A base is a proton acceptor.

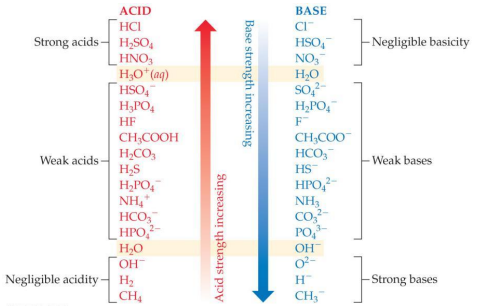
-Water is **amphiprotic(adj.)两性的**

-Conjugate(v.)结合

-conjugate bases 共轭碱

Acids above the line with H2O as a base are strong acids; Bases...

The substances between the lines with H2O are conjugate acid–base pairs in water.



-Negligible (adj.)微不足道的

-Autoionization 自电离 of Water

$H_2O(l) + H_2O(l) \rightleftharpoons OH^-(aq) + H_3O^+(aq)$

K_w: the ion product constant for water

$K_w = (K_c =) [H_3O^+][OH^-]$

At 25℃, Kw = 1.0×10^{-14}

- pH = -log[H^+]

Some other "p" systems are
➢pOH: –log[OH^-]
➢pK_w: –log K_w

(!!!此处对数底数为 10

$pH + pOH = pK_w = 14.00$

-litmus paper 石蕊纸, are used for less accurate measurements;

an **indicator 指示器** is one color in its acid form and another color in its basic form.

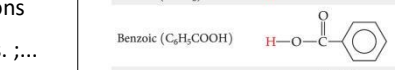
pH meters pH 计 are used for accurate measurement of pH;

electrodes 电极 indicate small changes in voltage to detect pH.

-Strong Bases 强碱: the alkali metal and heavier alkaline earth metal hydroxides (Ca2+, Sr^{2+}, and Ba2+)

-Weak Acids 弱酸 $K_a = [H_3O^+][A^-] / [HA]$

Table 16.2 Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous ($HClO_2$)	H—O—Cl—O	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	H—F	F^-	6.8×10^{-4}
Nitrous (HNO_2)	H—O—N=O	NO_2^-	4.5×10^{-4}
Benzoic (C_6H_5COOH)		$C_6H_5COO^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	H—O—Cl	OCl^-	3.0×10^{-8}
Hydrocyanic (HCN)	H—C≡N	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$C_6H_5O^-$	1.3×10^{-10}

*The proton that ionizes is shown in red.

-dissociate (v.)解离

电离程度→difference in conductivity and in rates of chemical reactions.

Strong acids: **Sulphuric acid**, H2SO4

Hydrochloric acid, HCl; **Nitric acid**, HNO3

From acidic oxides: SO2, SO3 , NOx

Weak acids: Partially ionized in solution,

Hydrofluoric acid , HF; **Acetic acid** CH3COOH

Formic acid HCOOH; **Benzoic acid** C6H5COOH

Carbonic acid H2CO3; **Phosphoric acid**, H3PO4

From: CO2 ,P2O3, P2O5

versus sulfurous acid 亚硫酸

-Polyprotic Acids 多质子酸

successive (adj.)逐次的，连续的

If the factor in the K_a values for the first and second dissociation has a difference of 3 or greater, the pH generally depends only on the first dissociation.

$\text{Percent ionization} = \frac{[H_3O^+]_{eq}}{[HA]_{initial}} \times 100$

-the base dissociation constant 碱解离常数

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq) \quad K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$

Table 16.4 Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid
Ammonia (NH_3)		NH_4^+
Pyridine (C_5H_5N)		$C_5H_5NH^+$
Hydroxylamine ($HONH_2$)		$HONH_3^+$
Methylamine (CH_3NH_2)		$CH_3NH_3^+$
Hydrosulfide ion (HS^-)		H_2S
Carbonate ion (CO_3^{2-})		HCO_3^-
Hypochlorite ion (ClO^-)		HClO

*The atom that accepts the proton is shown in blue.

弱碱的种类

1. Neutral substances

with an Atom that has a nonbonding pair of electrons

2. Anions of weak acids

-$K_a \times K_b = K_w$

-Hydrolysis 水解; Cation 阳离子; Anion 阴离子

-Hydrated Cations 水合阳离子

Transition and post-transition metals form hydrated cations

The water attached to the metal

is more acidic than free water molecules

↓使得

the hydrated ions are acidic.

-Salt Solutions 盐溶液 ~酸碱性判断

Group I/II metal cation + anion of a strong acid
→Neutral
Group I/II metal cation + anion of a weak acid
→Basic (like the anion)
Transition/Post-transition metal cation/polyatomic cation + anion of a strong acid
→Acidic (like the cation)
Transition/Post-transition metal cation/polyatomic cation + anion of a weak acid
compare Ka and Kb; whichever is greater dictates what the salt is.

- H—A 键越弱↓，酸的酸性越强

阴离子稳定性越高↑，酸的酸性越强↑

-Oxyacids (p.l.)含氧酸

非金属元素的电负性↑，具有相同结构的酸的酸性↑

如果一种元素可以形成一个以上的含氧酸，则 具有更多氧(O)原子的氧酸 的酸性更强

= If the oxidation number increases, the acidity increases. 氧化数↑，酸性增强↑

-Carboxylic Acids 羧酸: 其共轭碱(**羧酸阴离子**)**具有共轭形式**以稳定阴离子。

-所有 Brønsted-Lowry 酸和碱 也称为 Lewis 酸和碱

-Natural Indicators 自然的指示剂

Red cabbage 红甘蓝 boiled in water yields **dyes(p.l.)染料**

Hydrangeas(p.l.)绣球花

Ch24 有机化学

-Molecules with long nonpolar regions and polar regions act as surfactants, bringing polar material into aqueous solution (used in detergents and soaps).

-Carboxylic acids 羧酸

(—COOH) the most common acids

-Amines 胺 (—NH2, —NHR, or —NR2)

-Hydrocarbons 碳氢化合物

four basic types

Alkanes 烷烃, Alkenes 烯烃, Alkynes 炔烃, Aromatic hydrocarbons 芳烃

-Properties Common to Hydrocarbons

Are insoluble in water

Are nonpolar ↗↘

Are soluble in nonpolar solvents

Dispersion forces

↓ determine

Melting points and boiling points

-structural isomer 结构异构体

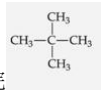
Def.: Compounds with the same molecular formula but different connections of atoms

Butane 丁烷

2-Methylpropane 2-甲基丙烷

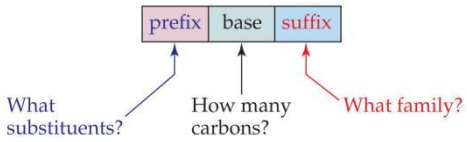
Pentane 戊烷

2-Methylbutane 2-甲基丁烷

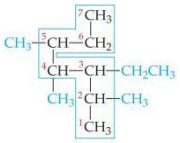


2,2-Dimethylpropane 2,2-二甲基丙烷

-Systematic Nomenclature



-alphabetically(adv.)按字母顺序



3-Ethyl-2,4,5-trimethylheptane

-Cycloalkanes 环烷烃

do not react with acids, bases, or **oxidizing agents 氧化剂**

the most important reaction is **combustion 燃烧**

2 C2H6(g) + 7 O2(g) → 4 CO2(g) + 6 H2O(l)

(exothermic, ΔH° = –2885 kJ)

-Cyclohexane 环己烷, Cyclopentane 环戊烷, Cyclopropane 环丙烷

-vertex 顶点, vertices 顶点

-saturated hydrocarbons = alkanes

饱和碳氢化合物 = 烷烃

Unsaturated hydrocarbons are more reactive than saturated hydrocarbons

-烯烃 Alkenes 的命名

longest chain must include BOTH carbon atoms that share the double bond; end name in -ene; lowest number possible given to double-bond carbon atoms; isomers also indicated.

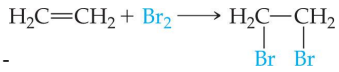
-Geometric Isomers 几何异构体

cis isomer 顺式异构体

trans isomer 反式异构体

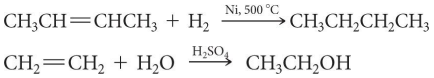
-炔烃 Alkynes 的命名

Naming: longest continuous chain containing both carbon atoms in the triple bond; name ends in -yne ; give C atoms in triple bond lowest number.



-

hydrogenation 氢化

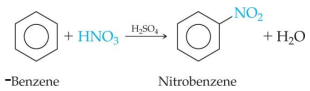


hydrogen halides 卤化氢

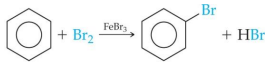
halogenation 卤化

-substitution reactions 取代反应

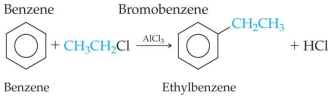
-additional reactions 加成反应



-Benzene Nitrobenzene



Benzene Bromobenzene



Benzene Ethylbenzene

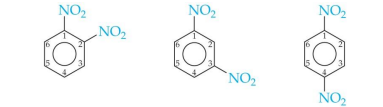
-Aromatic Nomenclature 芳香烃命名法

derivatives of benzene 苯的衍生物

1,2 = ortho- 邻位

1,3 = meta- 见位

1,4 = para- 对位



ortho-Dinitrobenzene 1,2-Dinitrobenzene mp 118 °C | *meta*-Dinitrobenzene 1,3-Dinitrobenzene mp 90 °C | *para*-Dinitrobenzene 1,4-Dinitrobenzene mp 174 °C

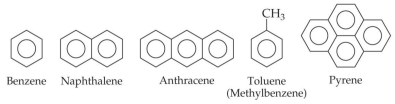
Benzene

Napthalene 萘

Anthracene 蒽

Toluene = Methylbeneze 甲苯

Pyrene 芘



Benzene Naphthalene Anthracene Toluene (Methylbenzene) Pyrene

-Functional Groups

portion 部分

Table 24.6 Common Functional Groups

Functional Group	Compound Type	Suffix or Prefix	Structural Formula
C=C	Alkene	*-ene*	H₂C=CH₂
—C≡C—	Alkyne	*-yne*	H—C≡C—H
—C—O—H	Alcohol	*-ol*	CH₃OH
—C—O—C—	Ether	*ether*	CH₃OCH₃
—C—X (X = halogen)	Alkyl halide or haloalkane	*-ide*	CH₃Cl
—C—N—	Amine	*-amine*	CH₃CH₂NH₂
—C(=O)—H	Aldehyde	*-al*	CH₃CHO
—C(=O)—C—	Ketone	*-one*	CH₃COCH₃
—C(=O)—O—H	Carboxylic acid	*-oic acid*	CH₃COOH
—C(=O)—O—C—	Ester	*-oate*	CH₃COOCH₃
—C(=O)—N—	Amide	*-amide*	CH₃CONH₂

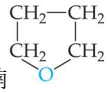
-the alcohol group 醇基 = the hydroxyl group 羟基

-醇以母烃命名;后缀更改为-ol

-Glycerol 甘油 -Phenol 苯酚 -Cholesterol 胆固醇

-Ether 醚 -dehydration 脱水

-diethyl ether 乙醚 $CH_3CH_2—O—CH_2CH_3$

-tetraydrifuran(THF)四呋喃 

-The carbonyl group 羰基 $C{=}O$

Aldehydes 醛, Ketones 酮,

Carboxylic acids 羧酸, Esters 酯, Amides 酰胺类

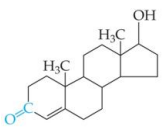
-methanal = formaldehyde 甲醛

-ethanal = acetaldehyde 乙醛

-propanone = acetone 丙酮

- 2-butnone = methyl ethyl ketone

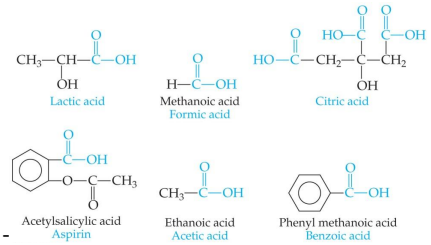
2-丁酮 $CH_3—C(=O)—CH_2CH_3$



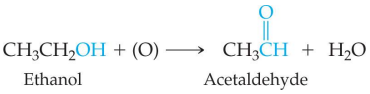
-testosterone 睾酮

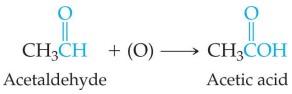
-除水外最重要的溶剂是丙酮，它溶于水并溶

解许多有机化合物。



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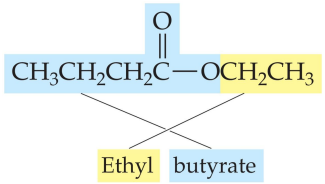




-酯的命名

Naming: name the alcohol part as an alkyl name;

separate word is the acid part as an -ate anion.



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